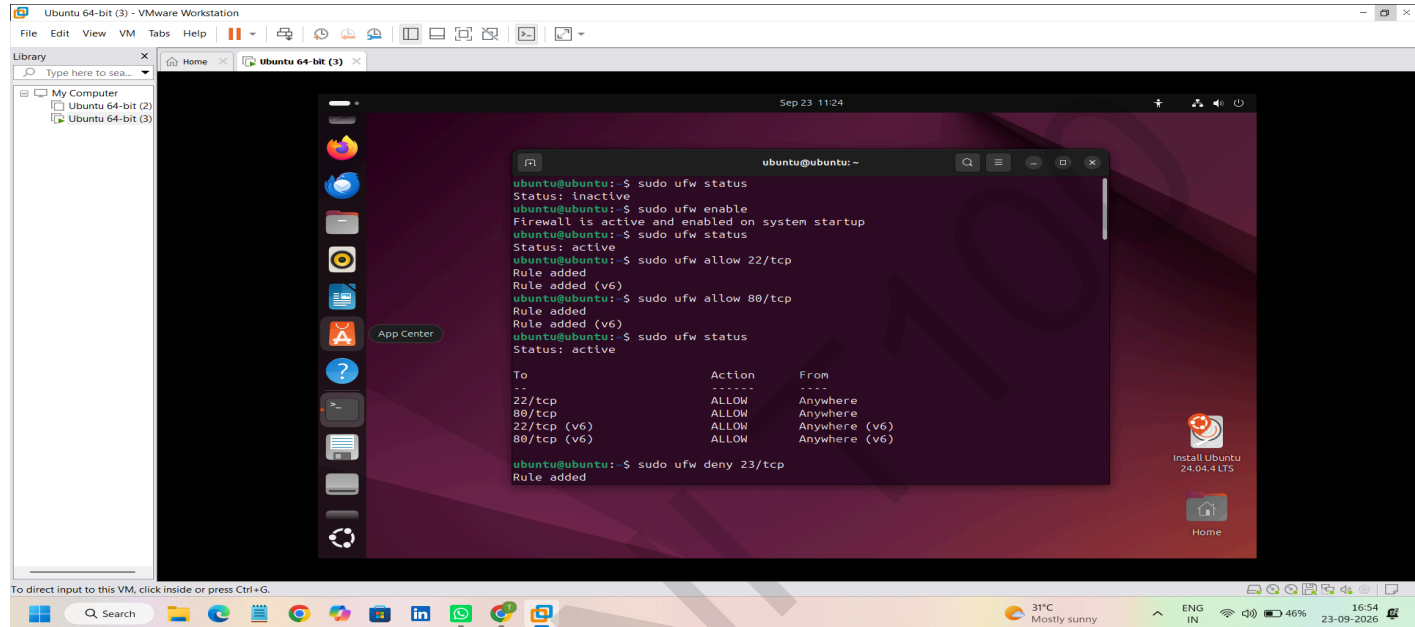


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PRACTICAL NO :- 10

Aim:- Configure and test firewall rules to control network traffic, filter packets based on specified criteria, and protect network resources from unauthorized access.

Step 1: Open Ubuntu Terminal

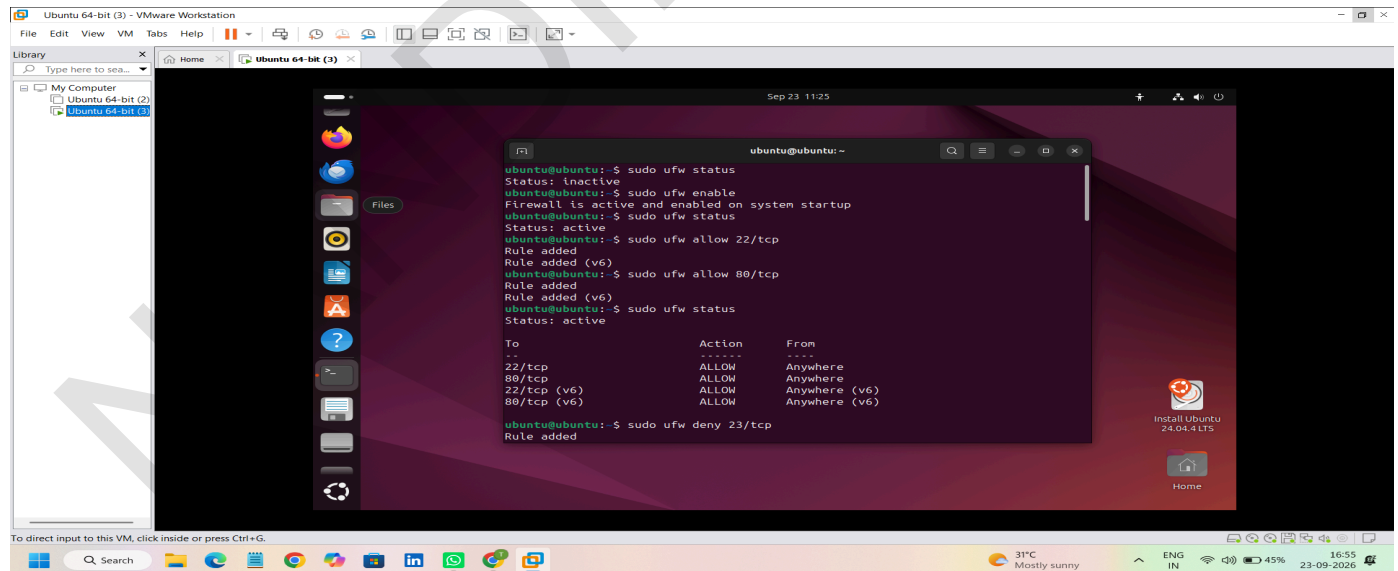


```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ sudo ufw status  
Status: inactive  
ubuntu@ubuntu:~$ sudo ufw enable  
Firewall is active and enabled on system startup  
ubuntu@ubuntu:~$ sudo ufw status  
Status: active  
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp  
Rule added  
Rule added (v6)  
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp  
Rule added  
Rule added (v6)  
ubuntu@ubuntu:~$ sudo ufw status  
Status: active  


| To          | Action | From          |
|-------------|--------|---------------|
| 22/tcp      | ALLOW  | Anywhere      |
| 80/tcp      | ALLOW  | Anywhere      |
| 22/tcp (v6) | ALLOW  | Anywhere (v6) |
| 80/tcp (v6) | ALLOW  | Anywhere (v6) |

  
ubuntu@ubuntu:~$ sudo ufw deny 23/tcp  
Rule added
```

Step 2: Enable the Firewall



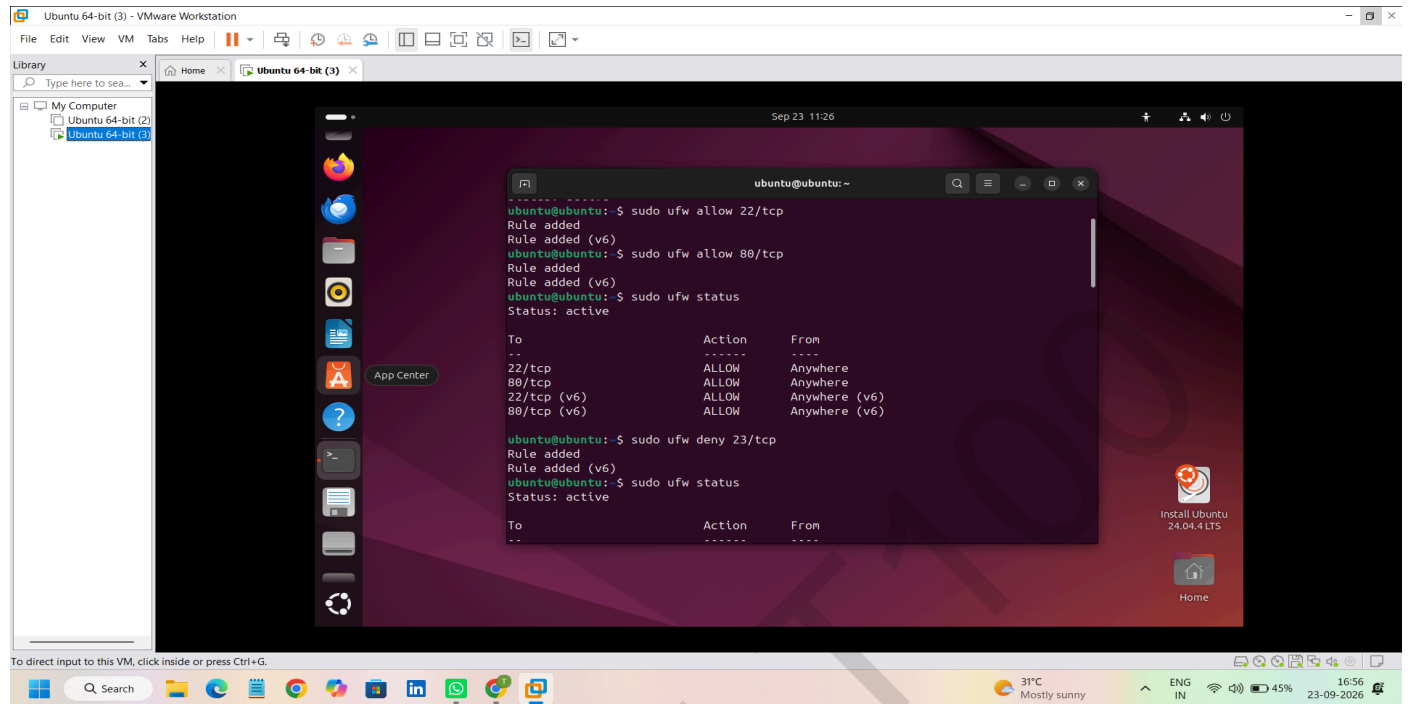
```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ sudo ufw status  
Status: inactive  
ubuntu@ubuntu:~$ sudo ufw enable  
Firewall is active and enabled on system startup  
ubuntu@ubuntu:~$ sudo ufw status  
Status: active  
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp  
Rule added  
Rule added (v6)  
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp  
Rule added  
Rule added (v6)  
ubuntu@ubuntu:~$ sudo ufw status  
Status: active  


| To          | Action | From          |
|-------------|--------|---------------|
| 22/tcp      | ALLOW  | Anywhere      |
| 80/tcp      | ALLOW  | Anywhere      |
| 22/tcp (v6) | ALLOW  | Anywhere (v6) |
| 80/tcp (v6) | ALLOW  | Anywhere (v6) |

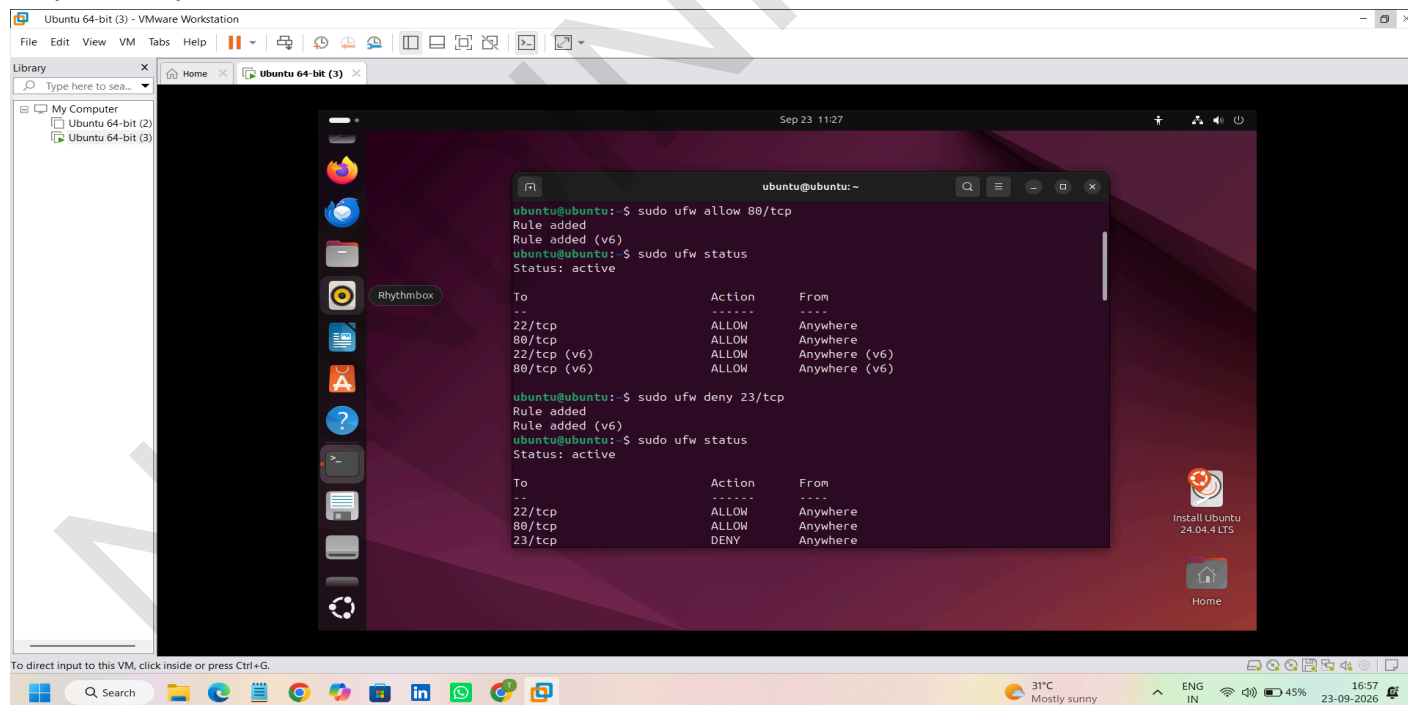
  
ubuntu@ubuntu:~$ sudo ufw deny 23/tcp  
Rule added
```

Step 3: Allow SSH Traffic

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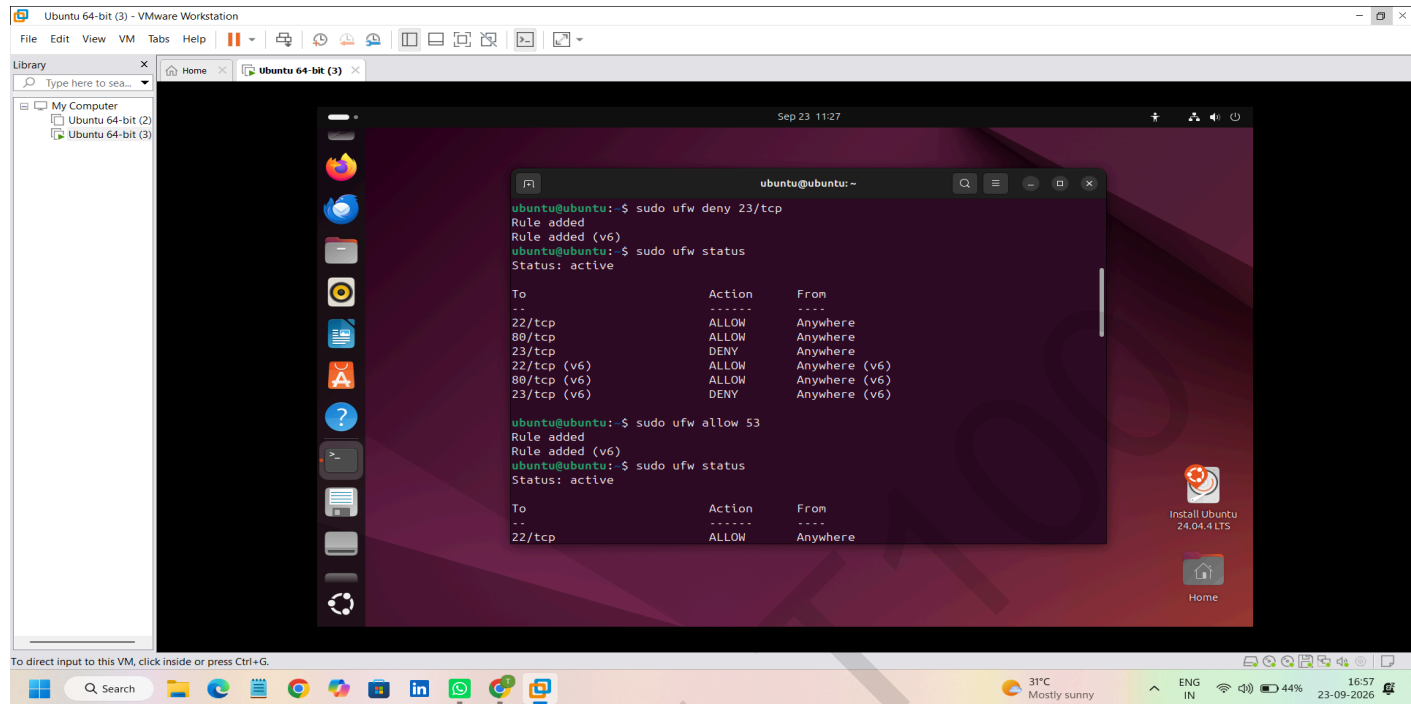
Step 4:- Step 4: Allow HTTP Traffic



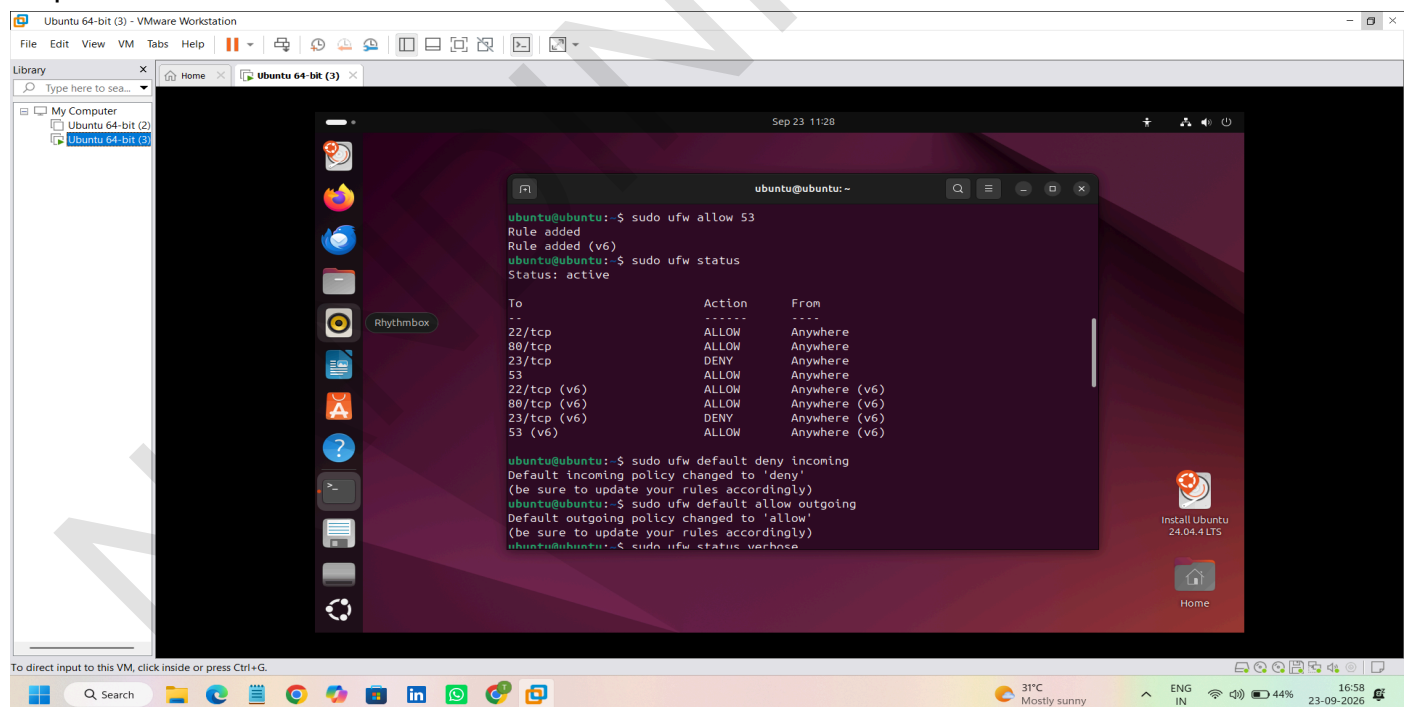
Step 5: Block a Port

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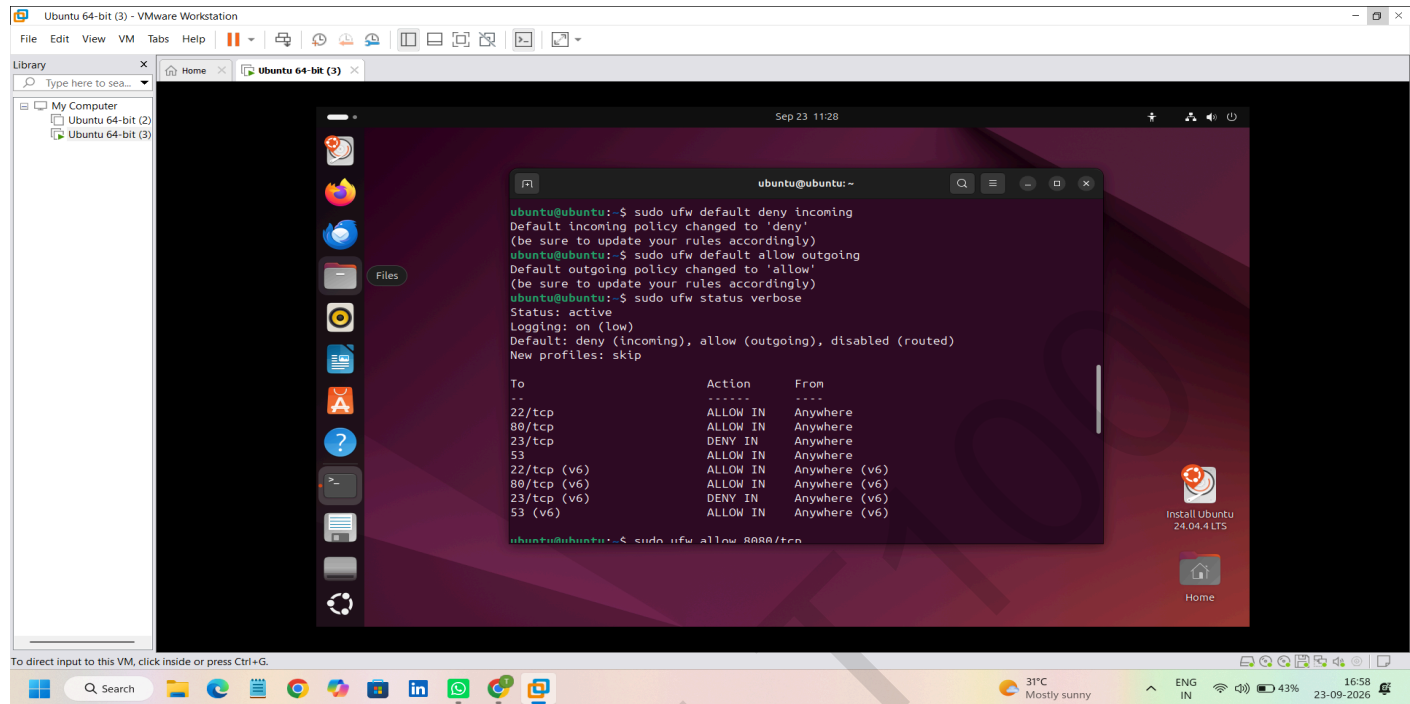


Step 6: Allow DNS Traffic

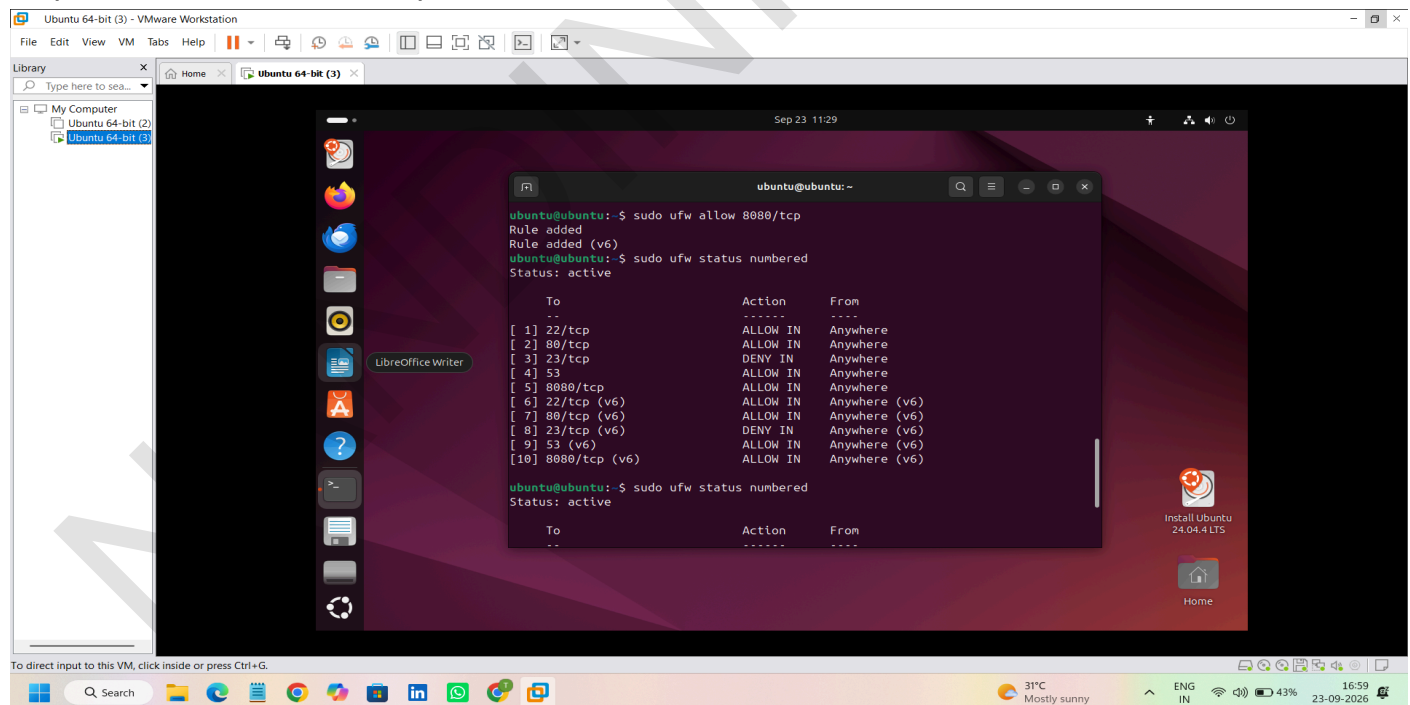


Step 7: Set Default Firewall Policies

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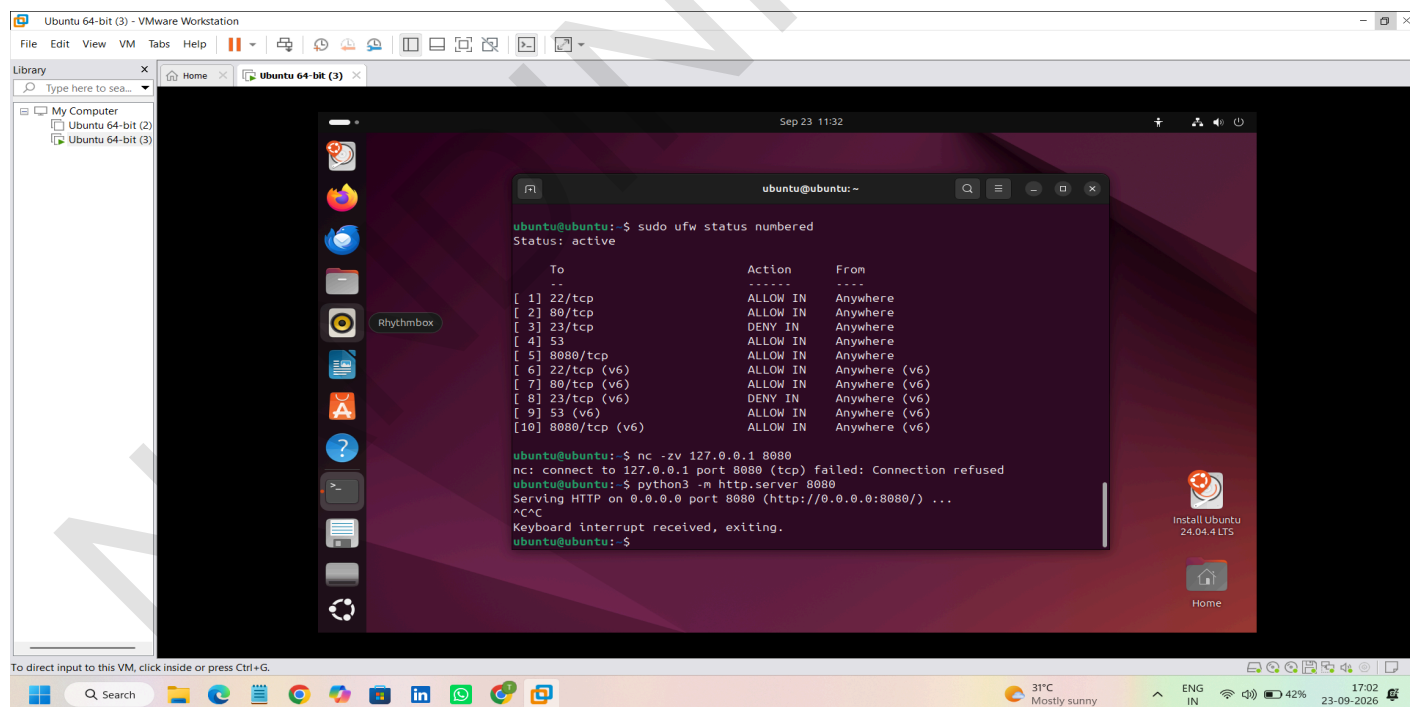
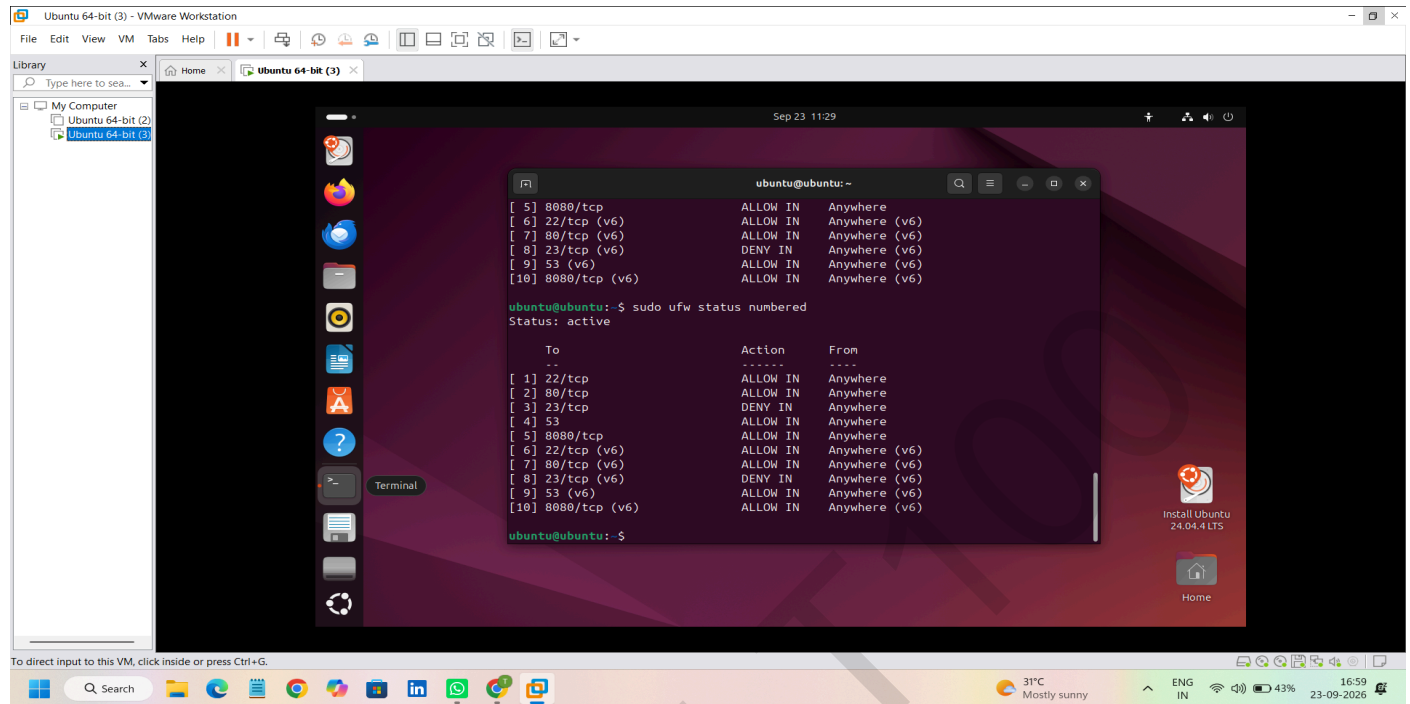


Step 8: Create a Rule for a Specific Port



Step 9: Test the Firewall

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Step 10: Test Blocking

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```
ubuntu@ubuntu:~  
ubuntu@ubuntu:~$ python3 -m http.server 8080  
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...  
^C^C  
Keyboard interrupt received, exiting.  
ubuntu@ubuntu:~$ sudo ufw deny 8080/tcp  
Rule updated  
Rule updated (v6)  
ubuntu@ubuntu:~$ sudo ufw status numbered  
Status: active  


| To                 | Action   | From          |
|--------------------|----------|---------------|
| --                 | ----     | ----          |
| [ 1] 22/tcp        | ALLOW IN | Anywhere      |
| [ 2] 80/tcp        | ALLOW IN | Anywhere      |
| [ 3] 23/tcp        | DENY IN  | Anywhere      |
| [ 4] 53            | ALLOW IN | Anywhere      |
| [ 5] 8080/tcp      | DENY IN  | Anywhere      |
| [ 6] 22/tcp (v6)   | ALLOW IN | Anywhere (v6) |
| [ 7] 80/tcp (v6)   | ALLOW IN | Anywhere (v6) |
| [ 8] 23/tcp (v6)   | DENY IN  | Anywhere (v6) |
| [ 9] 53 (v6)       | ALLOW IN | Anywhere (v6) |
| [10] 8080/tcp (v6) | DENY IN  | Anywhere (v6) |

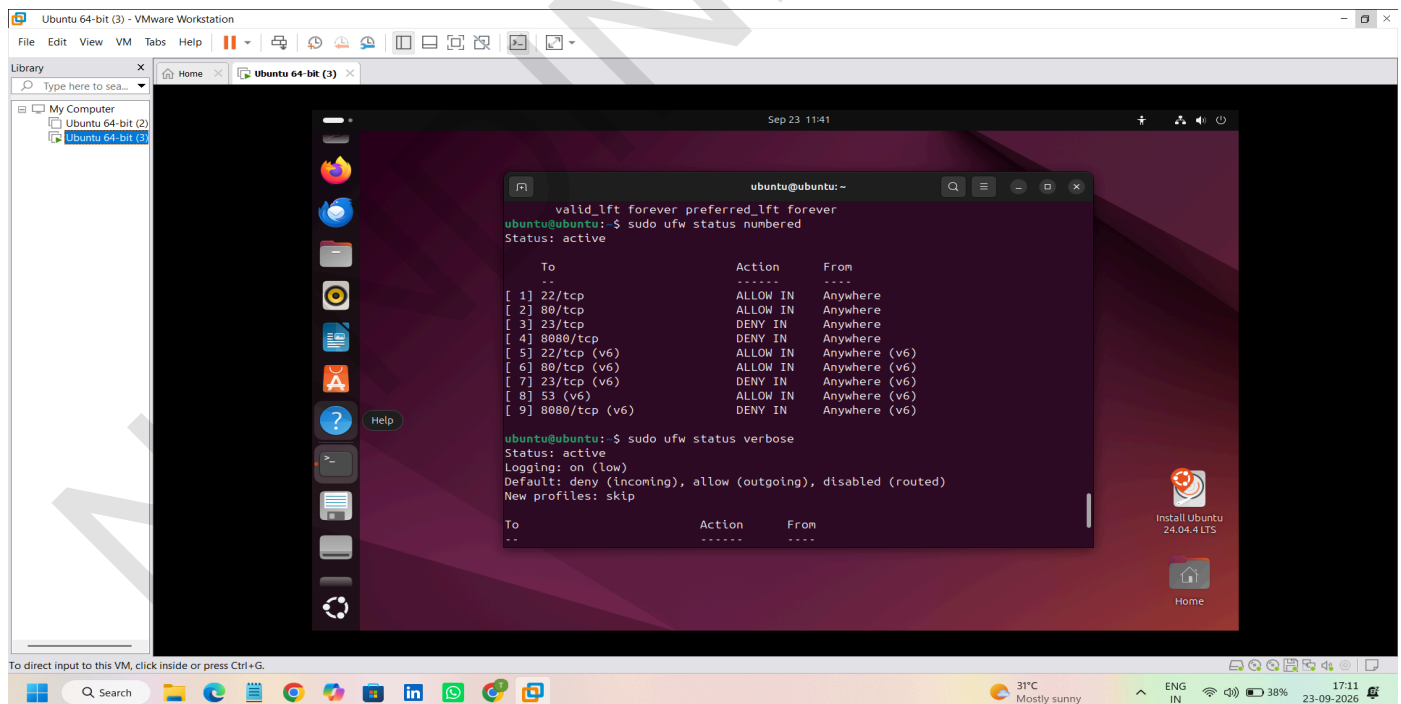
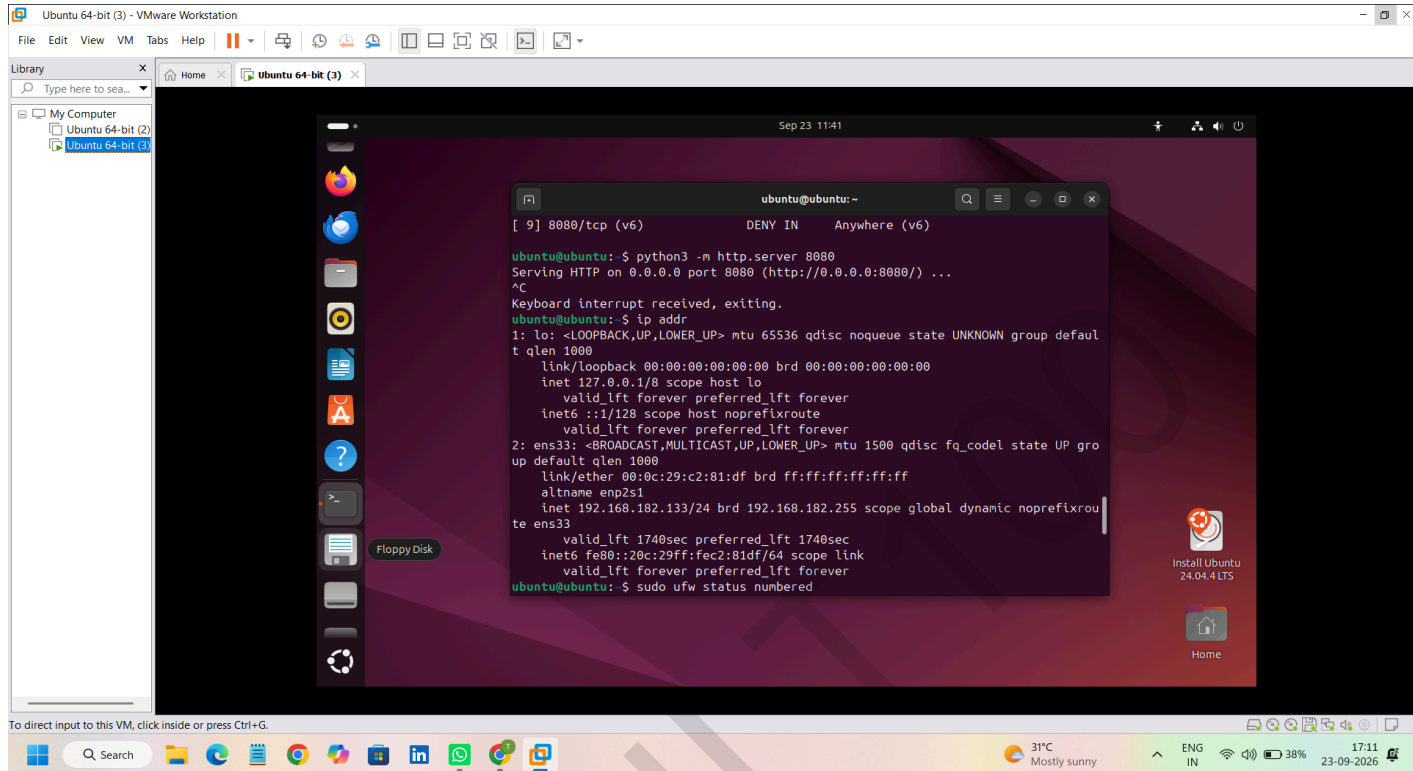
  
ubuntu@ubuntu:~$
```

```
ubuntu@ubuntu:~$ sudo ufw delete 4  
Deleting:  
allow 53  
Proceed with operation (y/n)? y  
Rule deleted  
ubuntu@ubuntu:~$ sudo ufw deny 8080/tcp  
Skipping adding existing rule  
Skipping adding existing rule (v6)  
ubuntu@ubuntu:~$ sudo ufw status numbered  
Status: active  


| To                 | Action   | From          |
|--------------------|----------|---------------|
| --                 | ----     | ----          |
| [ 1] 22/tcp        | ALLOW IN | Anywhere      |
| [ 2] 80/tcp        | ALLOW IN | Anywhere      |
| [ 3] 23/tcp        | DENY IN  | Anywhere      |
| [ 4] 8080/tcp      | DENY IN  | Anywhere      |
| [ 5] 22/tcp (v6)   | ALLOW IN | Anywhere (v6) |
| [ 6] 80/tcp (v6)   | ALLOW IN | Anywhere (v6) |
| [ 7] 23/tcp (v6)   | DENY IN  | Anywhere (v6) |
| [ 8] 53 (v6)       | ALLOW IN | Anywhere (v6) |
| [ 9] 8080/tcp (v6) | DENY IN  | Anywhere (v6) |

  
ubuntu@ubuntu:~$
```

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